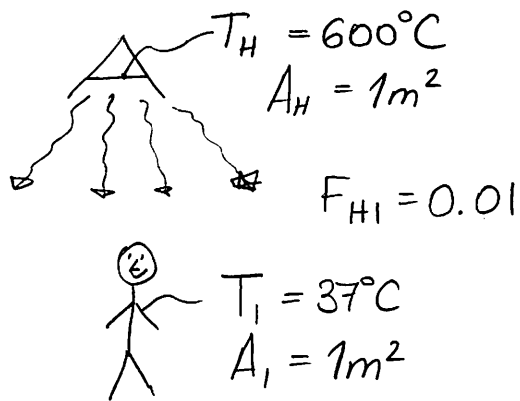


Example | Radiative heater. Find the heat flux to your body assuming:



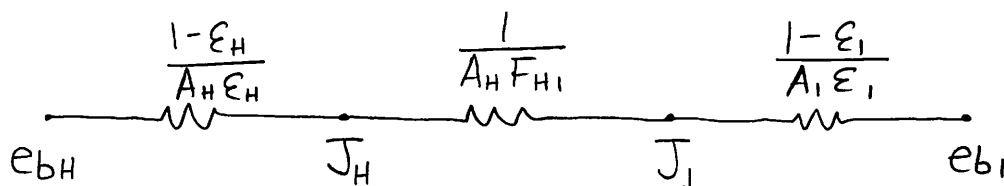
(a)  $\epsilon_H = 1.0$ ,  $\epsilon_I = 1.0$  (blackbody)

(b)  $\epsilon_H = 0.8$ ,  $\epsilon_I = 0.5$  (graybody)

(a) We can right away draw our thermal resistance diagram:

$$\Rightarrow q_{HI} = \frac{e_{bH} - e_{bI}}{\frac{1}{A_H F_{HI}}} = 320\text{W}$$

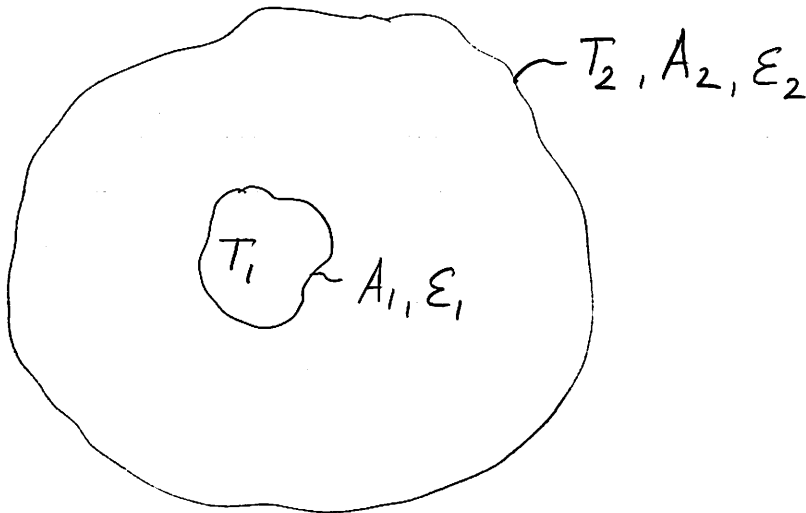
(b) Similar approach but more realistic:



$$q_{HI} = \frac{\overbrace{e_{bH} - e_{bI}}^{32431.65}}{\underbrace{\frac{1 - \epsilon_H}{A_H \epsilon_H}}_{0.25} + \underbrace{\frac{1}{A_H F_{HI}}}_{100} + \underbrace{\frac{1 - \epsilon_I}{A_I \epsilon_I}}_1} = 319\text{W}$$

This shows you the importance of relative contribution to resistances. In this case, view factor resistance dominates. (171)

## Gray Body Enclosures



Assuming that body 2 is much larger than body 1:

$$e_{b1} \quad \frac{1-\epsilon_1}{A_1\epsilon_1} \quad J_1 \quad \frac{1}{A_1F_{12}} \quad J_2 \quad \frac{1-\epsilon_2}{A_2\epsilon_2} \quad e_{b2}$$

$$q_{1 \rightarrow 2} = \frac{e_{b1} - e_{b2}}{\frac{1-\epsilon_1}{A_1\epsilon_1} + \frac{1}{A_1F_{12}} + \frac{1-\epsilon_2}{A_2\epsilon_2}} \Rightarrow \text{multiply by } \frac{A_1}{A_1}$$

$$q_{1 \rightarrow 2} = \frac{A_1(e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} - 1\right) + \frac{1}{F_{12}} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)}$$

Some special cases:

①  $A_2 \gg A_1$

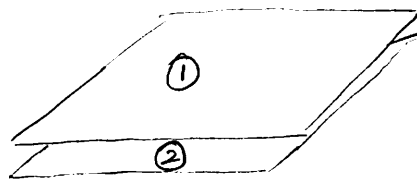
This means  $\frac{A_1}{A_2} \rightarrow 0$  and  $F_{12} = 1$

Back substituting into our solution

$$q_{1 \rightarrow 2} = \frac{A_1 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} - 1 + 1 + 0 \left( \frac{1}{\epsilon_2} - 1 \right)}$$

$$\boxed{q_{1 \rightarrow 2} = A_1 \epsilon_1 (e_{b1} - e_{b2})} \Rightarrow \text{Smaller body behaves like a black body with finite emissivity.}$$

② Parallel Plates (small gap between them)

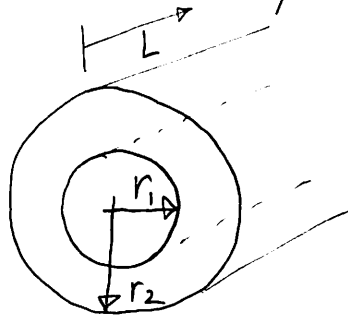


$$\frac{F_{12}}{A_1} = 1 = \frac{F_{21}}{A_2}$$

$$q_{1 \rightarrow 2} = \frac{A_1 (e_{b1} - e_{b2})}{\left( \frac{1}{\epsilon_1} - 1 \right) + \frac{1}{F_{12}} + \frac{A_1}{A_2} \left( \frac{1}{\epsilon_2} - 1 \right)} = \frac{A_1 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} - 1 + 1 + \frac{1}{\epsilon_2} - 1}$$

$$\boxed{q_{1 \rightarrow 2} = \frac{A_1 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}}$$

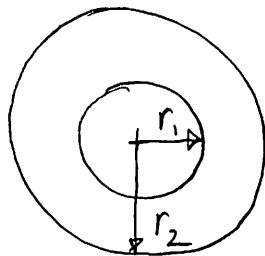
### ③ Concentric Cylinders (Length $L$ )



$$F_{12} = 1 ; \quad \left. \begin{array}{l} A_1 = 2\pi r_1 L \\ A_2 = 2\pi r_2 L \end{array} \right\} \frac{A_1}{A_2} = \frac{r_1}{r_2}$$

$$q_{1 \rightarrow 2} = \frac{2\pi r_1 L (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1}{r_2} \left( \frac{1}{\epsilon_2} - 1 \right)}$$

### ④ Concentric Spheres

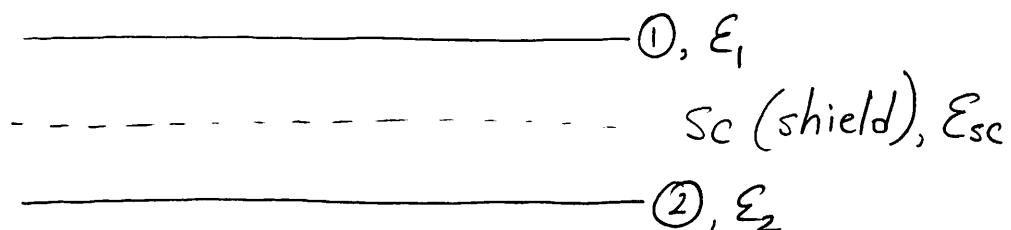


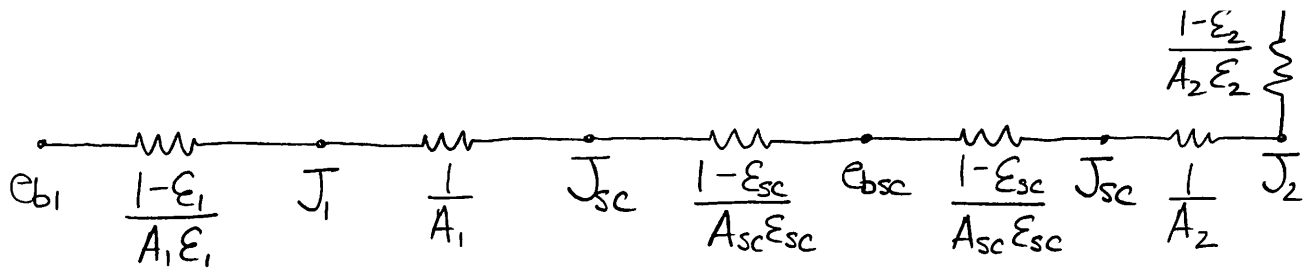
$$\frac{A_1}{A_2} = \frac{r_1^2}{r_2^2} ; \quad F_{12} = 1$$

$$q_{1 \rightarrow 2} = \frac{4\pi r_1^2 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1^2}{r_2^2} \left( \frac{1}{\epsilon_2} - 1 \right)}$$

### Radiation Shields

Used to reduce radiative heat loss  
For a flat plate arrangement:





Since  $A_1 = A_{sc} = A_2$  (parallel plates)  
 $F_{12} = F_{sc,2} = F_{2,sc} = 1$

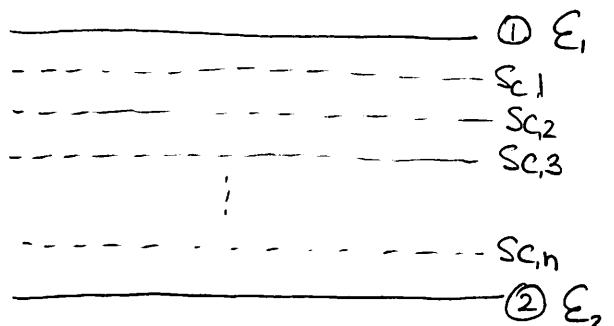
Our result simplifies to:

$$q_{12} = \frac{A (e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} - 1\right) + 1 + \left(\frac{1}{\epsilon_2} - 1\right) + 2\left(\frac{1}{\epsilon_{sc}} - 1\right) + 1}$$

$$q_{12} = \frac{A (e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_{sc}} - 1\right)} \Rightarrow \text{For 1 shield}$$

Note the solution is identical to the parallel plate solution from before except now we have an additional term in the denominator that diminishes  $q_{12}$ .

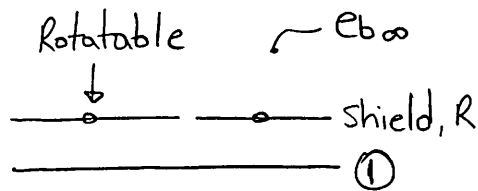
For  $n$  shields in series: (with identical radiative properties)



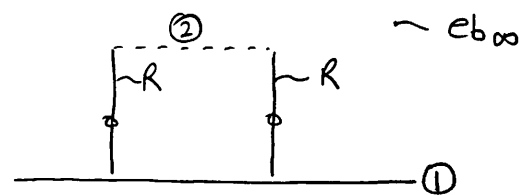
$$q_{12} = \frac{A (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} + \frac{2n}{\epsilon_{sc}} - (n+1)}$$

Example | Consider NASA's rotatable radiation shield shown below. Calculate its effectiveness in the on & off positions:  $\epsilon_{sc} = 0.1$ ,  $\epsilon_1 = 0.5$

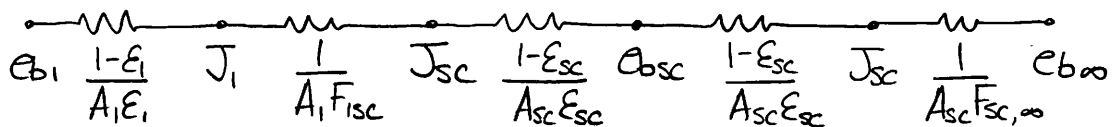
State I: ON



State II: OFF



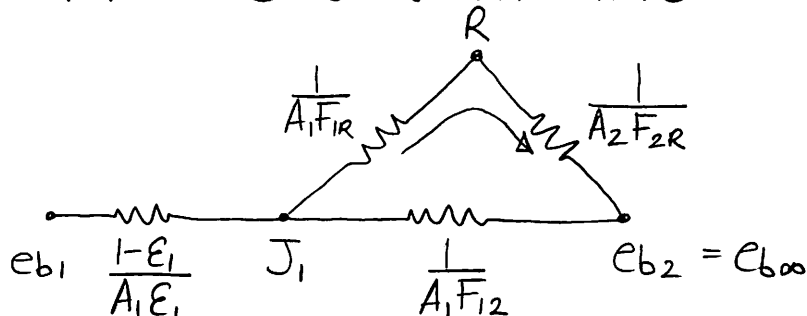
Lets solve the ON case first:



We know  $A_1 = A_{sc} = A$  (parallel infinite plates)  
 $F_{isc} = F_{sc2} = 1$

$$q_{1\infty} = \frac{A_1 (e_{b1} - e_{b\infty})}{\frac{1}{\epsilon_1} + \frac{2}{\epsilon_{sc}} - 1} \Rightarrow \text{State I, ON}$$

The OFF case is a bit more tricky to solve:



\* The shields in the OFF state act as adiabatic surfaces due to symmetry, so  $e_{bR} = J_R$

From our view factor tables:

$$F_{12} = \sqrt{2} - 1$$

$$F_{1R} = 2 - \sqrt{2}$$

$$q_{12} = \frac{A_1 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} - 1 + \sqrt{2}} \Rightarrow \text{State II, OFF}$$

Now we can calculate the ratio of heat transfer from the thermal resistance ratios (denominator):

$$\frac{R_I}{R_{II}} = \frac{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_{sc}} - 1}{\frac{1}{\epsilon_1} - 1 + \sqrt{2}} = 13.2$$

So when the shield is ON, it is 13.2 times better at limiting radiative heat transfer.

Note here the importance of choosing  $\epsilon_{sc}$ . If  $\epsilon_{sc} \ll 1$ , then radiation shield is much better performing. This is why scientists use highly reflective surfaces for radiation shielding like aluminum foil.